

Two Stage Ultra-Wideband Power Amplifier Based on GaN HEMT

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Abstract: The research of a two stage ultra- broadband power amplifier based on GaN HEMT which operates in the frequency ranging from 3 to 8 GHz, is presented in this paper. The power devices TGF2023-2-02 and TGF2023-2-10 based on 0.25 μ m GaN technology are respectively adopted in driver stage and power stage. A frequency compensation and multi-side impedance matching approach are employed in such a way that simultaneously provides appropriate output power and optimal wideband matching. And a fan shaped micro strip line is implemented in the input matching network of the driver stage to achieve broadband impedance transformer. It has been observed that 15-30 W output power with minimum small signal gain of 19.9 dB is demonstrated in the entire band.

Keywords: GaN HEMT; Ultra broadband; Power amplifier; Output power

1 Introduction

Design of ultra- broadband power amplifiers is an important problem in applications including electronic warfare (EW), instrumentation and soft ware-defined radio (SDR). The increasing request for broadband PAs in different system applications has pushed the research effort in the development of broader bandwidth and higher power active devices [1-2]. GaN on SiC HEMTs have particular capabilities such as great output power, good thermal conductivity, especially the very high drain-to-source breakdown voltage, resulting in higher device's optimum impedance, simplifying the output transformation ratio towards 50-ohm and allowing broadband design [3]. In recent years, based on the GaN HEMTs, numerous amplifiers have been developed by many researchers [4-8]. And to the best of the authors' knowledge, the maximum RF output power is less than 10 W over frequencies ranging from 3-8 GHz. More distributed and balanced structures are adopted to achieve higher power and broader bandwidth, which both need at least two active devices and have a strong demand on the consistency of the power devices, simultaneously having a higher power consumption.

This paper proposes a two stage ultra-wideband amplifier based on GaN HEMT. A frequency compensation and multi impedance broadband matching approach are implemented to achieve the wideband features including higher gain and output power, especially a fan shaped strip line with broadband characteristics is used in the input matching network of

the driver stage. The actual results demonstrate the availability and accuracy of the design approach. Good stability, high gain and broad bandwidth have been attained. The proposed design offers more than 19.9 dB power gain over 3–8 GHz and that the output power is 15-30 W.

2 Broadband power amplifier design

The power devices TGF2023-2-02 and TGF2023-2-10 from TriQuint Corporation are chosen to design the proposed wideband amplifier in Class-AB mode, which are respectively a 12 W and 50 W transistors capable to be used within DC – 18 GHz. The wideband amplifier is designed in the software ADS and its schematic diagram is shown in Figure 1. This circuit consists of two main sections: driver and power stages, which are individually designed, then are linked to each other. The elements of the inter stage need to be optimized to drive accurately the last stage and to keep the power gain at the maximum values in the whole band. As concerns the driver stage, it's designed mainly to improve the gain of the amplifier. And the goal of the power stage is to provide high output power. The two stages both consist of a matching circuit, DC-biased circuit and GaN HEMT. Moreover, substrate materials should be carefully chosen considering trade-off between the circuit size and the electrical performance. The DC bias and matching networks are realized with Rogers 4350 B (higher dielectric constant of 3.66) which reduce the size of the matching circuits and a substrate thickness of 0.254 mm. Both the input and output ports are connected to the standard 50 Ω load.

To optimize the design performance, several simulation analyses were performed in linear and non-linear RF mode. The output matching network of power amplifier is designed according to optimal impedance of the load-pull based on the model of the RF power transistor build before to guarantee optimal performance under large signal operation [9]. Based on these values, the optimum matching at frequency of 8 GHz for frequency compensation and multi-side impedance matching approach are applied for the broadband matching. Transmission line elements and capacitors in this matching network are needed to be tuned to reach optimized load matching in the whole frequency. In addition, an important effort for input matching of the driver stage was required to control the gain flatness over 3 to 8 GHz. An open radial stub stub1 with the characteristic of broadband impedance transformer is

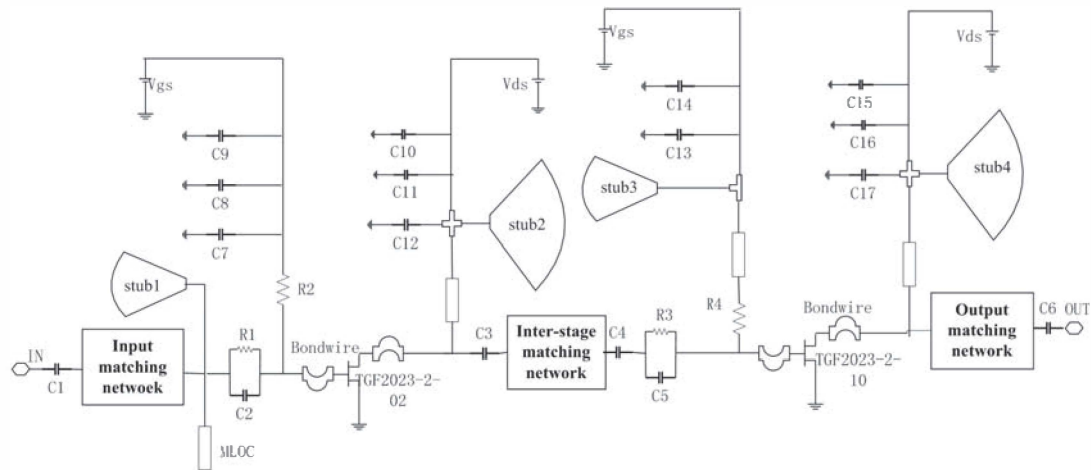


Figure 1 Schematic diagram of the proposed power amplifier

used during the multi-side impedance matching design. Matching networks are connected to the gate and drain pad of transistors with 25 μm gold wire bonds.

Bias circuit is designed to provide all the required bias voltage and bias current of the proposed amplifier. The two stages are both designed under DC bias of $V_{ds}=28\text{ V}$, $V_{gs}=-3\text{ V}$ and I_{ds} is respectively 125 mA and 500 mA. Inaccurate design of a DC bias network may lead to oscillation if the RF signal leaks to the DC supply. Design of bias network by using a high impedance quarter wave length transformer and radial stub shown as stub2, stub3 and stub4 is very common [2]. They are opened for the odd harmonic waves and closed to the even harmonic waves, which is useful to prevent the leakage of energy from the bias network and promote the linearity and output power of the amplifier. R1, R2, R3 and R4 are used to achieve required stability within the band of interest. But improved stability and better VSWR are achieved at a cost of gain. Therefore, to compensate the gain, a parallel capacitor C2 and C5 is used along with the series resistor R1 and R3. This arrangement is called hybrid stability network which makes it possible to attain stability and broadband matching simultaneously for R1 and C2 respectively tuning the gain of low and high frequency signal in the operating band, so is R3 and C5. The resistor R2 and R4 between DC power supply and the transistor gate for the GaN HEMT neither consumes any power nor produces any heat, because the gate current is nearly zero under ideal conditions. The input and output coupling capacitors are shown as C1, C3, C4 and C6 respectively, which can insulate the DC current and match simultaneously.

3 Results and discussion

To have a reliable design, simulations of all transmission lines are validated with electromagnetic simulator (momentum ADS) to take the coupling and discontinuity effects into account. And in the proposed power amplifier, many holes are made on the PCB to avoid cavity resonance effects. Figure 2 shows the momentum ADS layout of the proposed power amplifier.

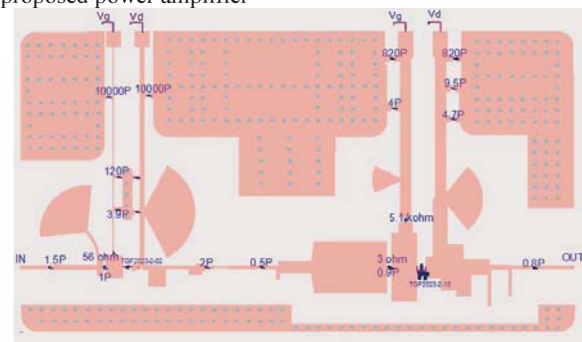


Figure 2 The momentum ADS layout

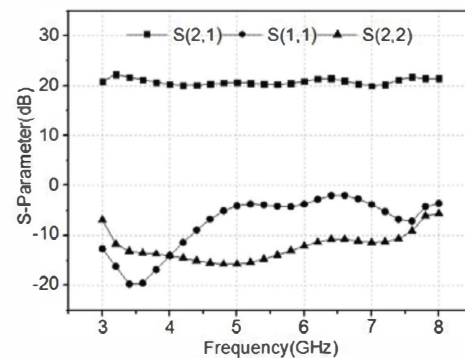


Figure 3 Simulated S-parameter of the circuit

Figure 3 shows the S-parameter simulation results of the circuit. A power gain $S(2, 1)$ of more than 19.9 dB in the frequency range of 3-8 GHz is obtained. Input return loss $S(1, 1)$ is a little poor for higher frequencies from 5-7 GHz. And except that, $S(1, 1)$ is less than -5 dB. Output return loss $S(2, 2)$ shows great performance and -6.5 to -16 dB is attained from 3 to 8 GHz. Large signal simulation is performed when the input CW signal is applied. Figure 4 shows the output power and power gain versus input power at the frequency of 5 GHz. It shows that the saturated output power is 44.6 dBm. The simulated results of the saturated output power, power gain, and PAE versus input frequency are drawn in Figure 5. It can be seen that the proposed design offers 15-30 W output power with minimum power gain of 10 dB and more than 16.5 % power added efficiency (PAE) over 3-8 GHz. The PAE has a larger variation, the main

reason for which is that PAE is expressed based on output power (P_{out}) and input power (P_{in}) with Watts, as shown in below

$$PAE = \frac{P_{out}(W) - P_{in}(W)}{P_{DC1} + P_{DC2}} \quad (1)$$

whereas figure 5 gives the P_{out} with dBm. In general, the power difference is bigger when it is expressed with Watts than with equivalent dBm. P_{DC1} and P_{DC2} respectively represent the DC power supply of the driver and power stage. The sum of their current values varies largely with frequency too, which also leads to the larger variation of PAE. Table I shows a summary of the results from reported literature and our work. As it can be seen, the proposed design has a better performance including bandwidth and output power than others.

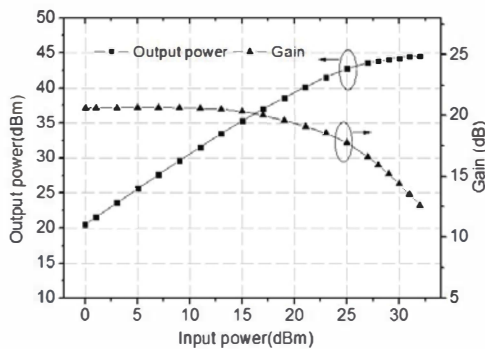


Figure 4 Output power and gain versus input power at 5 GHz

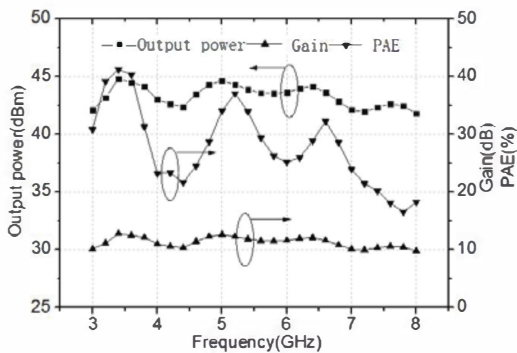


Figure 5 Saturated output power, power gain, and PAE versus frequency

Table I Comparison of characteristics

Ref.	Frequency (GHz)	Output power (dBm)	Transistor
[4]	1-3	45.8	GaN HEMT
[5]	1-3	37.7- 41.7	GaN HEMT
[6]	0.9-1.5	43.2	GaN HEMT
[7]	8	35.2	GaN HEMT
[8]	0.3-2.5	39	GaN HEMT
[10]	0.15-0.5	37	GaN HEMT
[11]	1-7	36	GaN HEMT
This work	3-8	41.7-44.7	GaN HEMT

4 Conclusions

A two stage ultra-wideband amplifier is designed using the wideband matching networks for frequency ranging from 3-8 GHz. 15-30 W output power is demonstrated in the whole band. Furthermore, the proposed PA shows a minimum small signal gain of 19.9 dB and good gain flatness in all frequency bandwidth. The PAE is higher than 16.5 % with a maximum of 42.9 %. These results demonstrate the effectiveness of the adopted amplifier design approach. All these performance have important reference value for the future development on ultra-wideband power amplifier.

References

- [1] Nam-Tae Kim. Design of an ultra-broadband power amplifier using distributed network synthesis. *IEICE Electronics Express*, 2013, 10(19): 1- 8.
- [2] Mehdi Forouzanfar, Rouhollah Fegghi, Mojtaba Joodaki. An 8.8 - 9.8 GHz 100W hybrid solid state power amplifier for high power applications. *ICEE*. 2014: 433-437.
- [3] G Mougnot, H Blanck, M Le Pipe, et al. Three Stage 6-18 GHz High Gain and High Power Amplifier based on GaN Technology. *IEEE*, 2010: 1392-1395.
- [4] C Berrached, D Bouw., M Camiade, et al. Wideband High Efficiency High Power GaN Amplifiers Using MIC and Quasi-MMIC Technologies. *The 43rd European Microwave Conference*. 2013: 1395-1398.
- [5] Xuekun Du, Zhenhai Shao, Changjiang You, et al. A New Method of Designing High Efficiency and Wideband Power Amplifier. *IEEE*, 2014: 638-641.
- [6] Sergio Di Falco, Antonio Raffo, Davide Resca, et al. GaN Power Amplifier Design Exploiting Wideband Large-Signal Matching. *IEEE*, 1999:277-279.
- [7] Wei Tao. Research on X band GaN Microwave Power mplifier. *Internet of Things Technologies*, 2012, (5): 61-63.
- [8] Xie Xiaofeng, Xiao Shiwei, Shen Chuan, et al. Design and Realization of a Multi-Octave GaN Distributed Power Amplifier. *Journal of microwaves*, 2013, 29(04):74-77.
- [9] Cheng Zhiqun, Jia Minshi, Luan Ya, et al. Solid-state wideband GaN HEMT power amplifier with a novel negative feedback structure. *Journal of conductors*, 2014, 35(12): 1-4.
- [10] F. A. Mughal, A. Kashif, N.B. Cheema. Design and Fabrication of a GaN HEMT Based Amplifier for Wideband Applications. *IEEE*, 2015: 554-556.
- [11] R. Giofrè, P. Colantonio, F. Giannini, et al. 1-7 GHz Single-Ended Power Amplifier based on GaN HEMT grown on Si-Substrate. *Proceedings of the 7th European Microwave Integrated Circuits Conference*, 2012: 425-428.